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Link Layer and LAN

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Unit – 4 Link Layer and LAN Roadmap



- Introduction
- Error detection, correction
- Multiple access protocols
- Switched LANs
 - addressing, ARP
 - Ethernet
 - switches
- A day in the life of a web request
- Physical layer
- Wireless LANs: IEEE 802.11

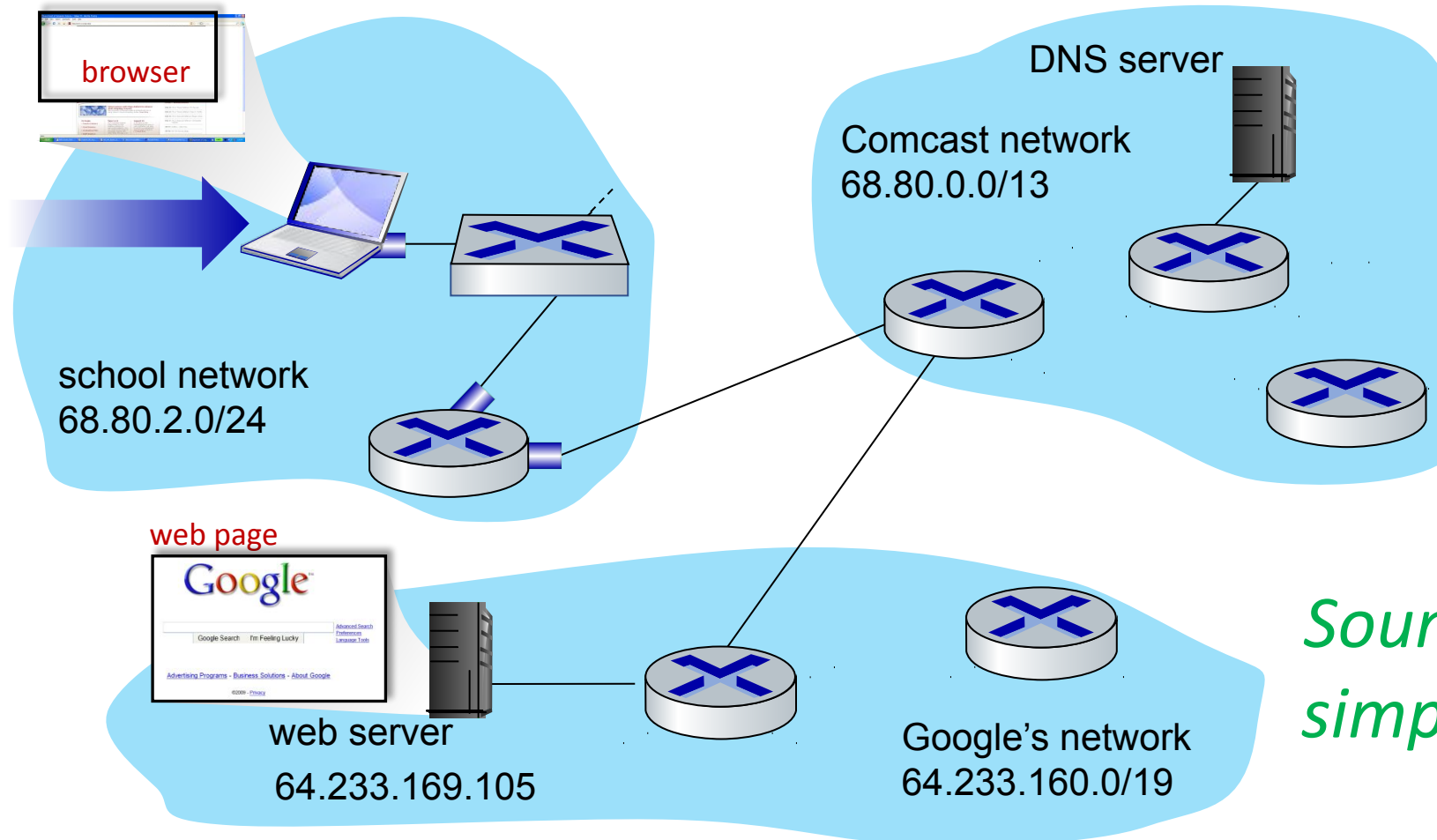
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Synthesis : A day in the life of a web request

- Our journey down the protocol stack is now complete!
 - application, transport, network, link
- Putting-it-all-together: synthesis!
 - *Goal*: identify, review, understand protocols (at all layers) involved in seemingly simple scenario: requesting www page
 - *Scenario*: student attaches laptop to campus network, requests/receives `www.google.com`

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A day in the life of a web request



Scenario:

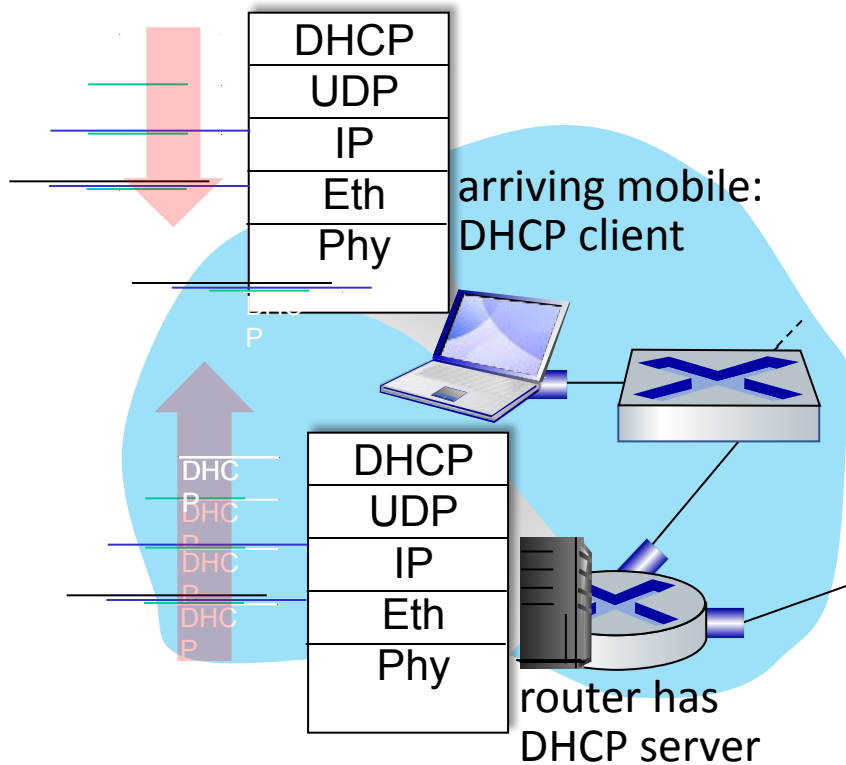
- Arriving mobile client attaches to network ...
- Requests web page: www.google.com

Sounds simple!



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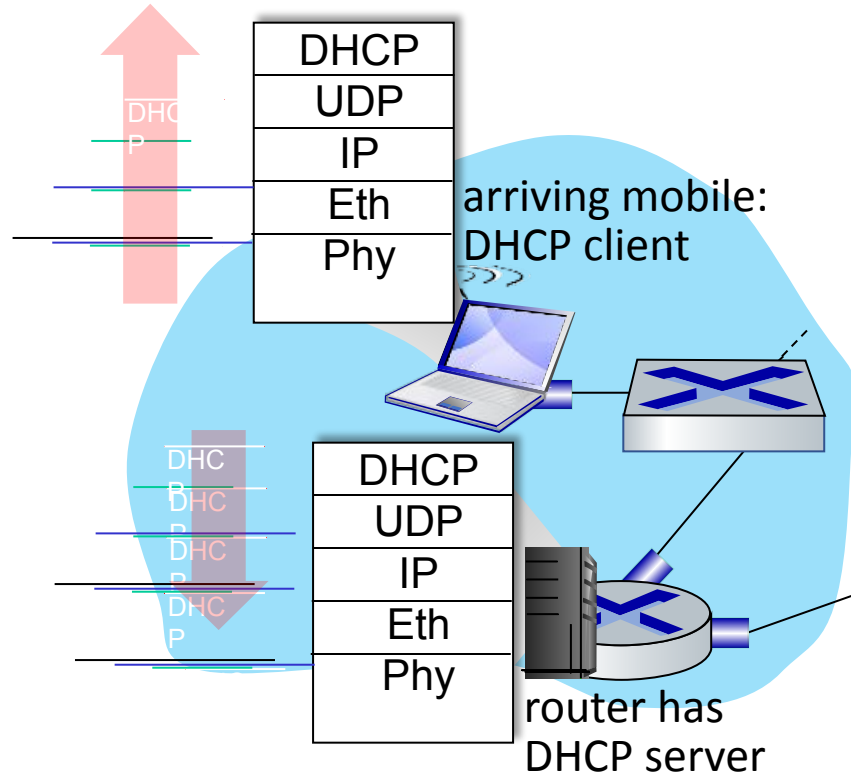
A day in the life of a web request



- Connecting laptop needs to get its own IP address, addr of first-hop router, addr of DNS server: use **DHCP**
- DHCP request **encapsulated** in **UDP**, encapsulated in **IP**, encapsulated in **802.3** Ethernet
- Ethernet frame **broadcast** (dest: FFFFFFFFFFFFFFFF) on LAN, received at router running **DHCP** server
- Ethernet **demuxed** to IP demuxed, UDP demuxed to DHCP

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A day in the life : Connecting to the Internet

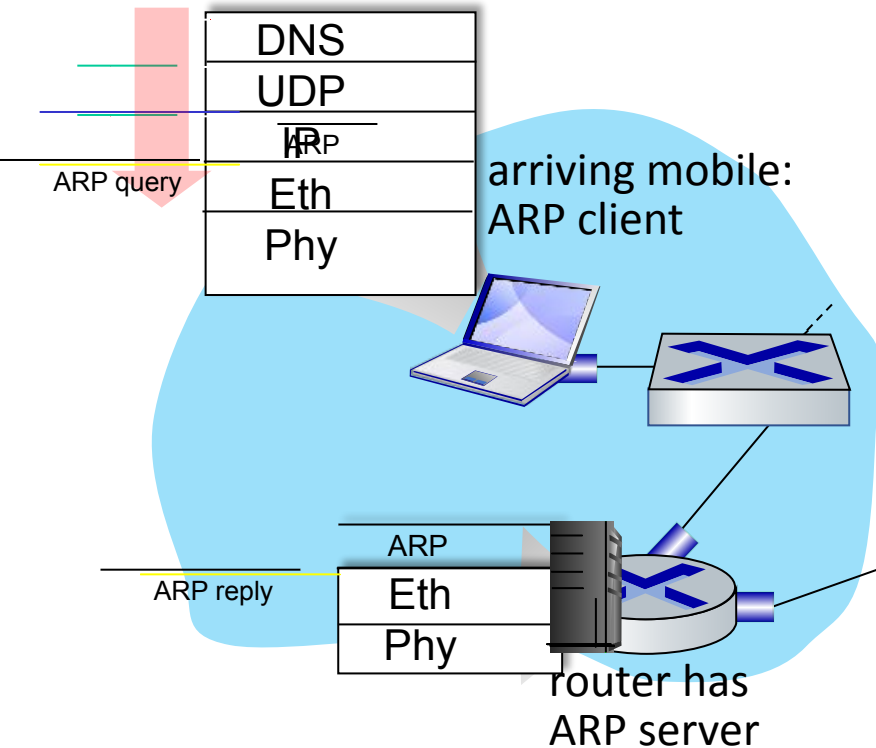


- DHCP server formulates **DHCP ACK** containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- Encapsulation at DHCP server, frame forwarded (**switch learning**) through LAN, demultiplexing at client
- DHCP client receives DHCP ACK reply

Client now has IP address, knows name & addr of DNS server, IP address of its first-hop router

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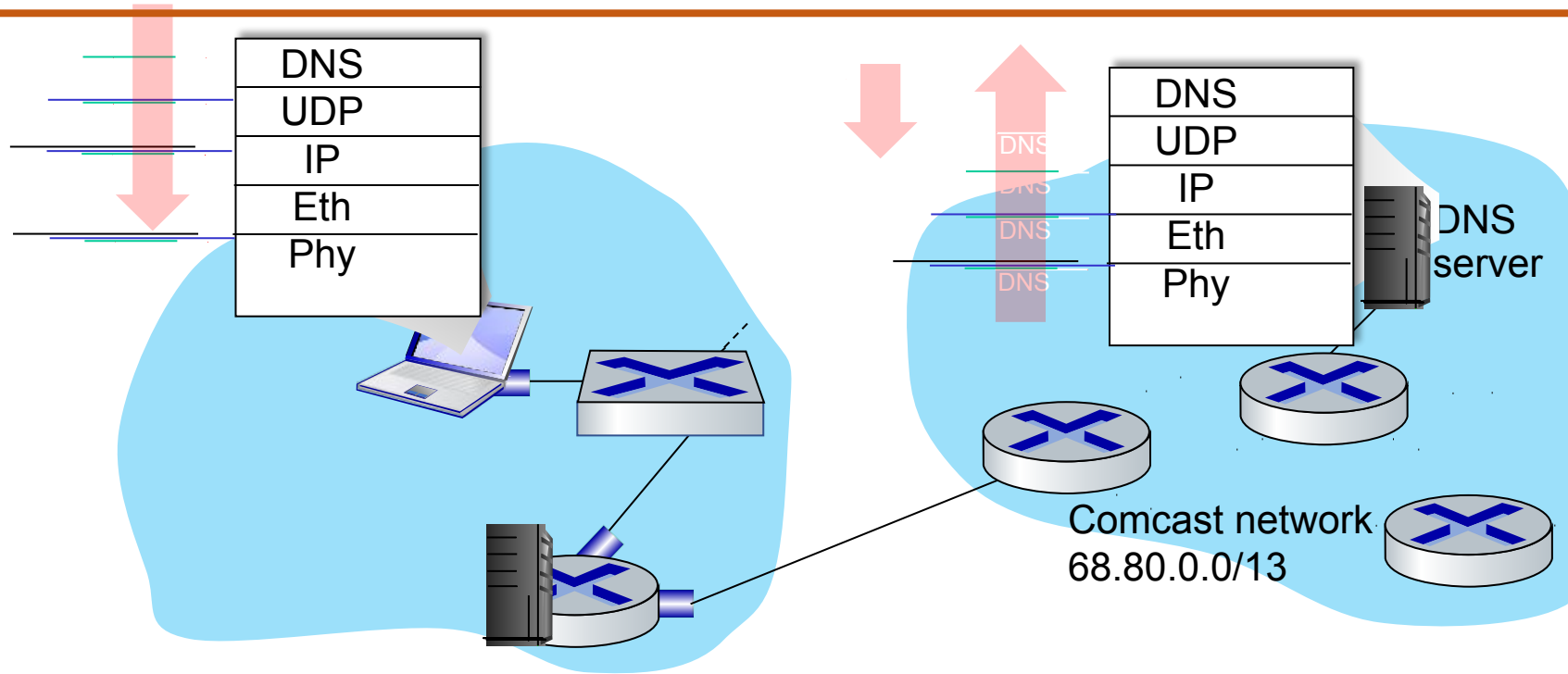
A day in the life.... ARP (Before DNS, Before HTTP)



- Before sending **HTTP** request, need IP address of **www.google.com**: **DNS**
 - DNS query created, encapsulated in UDP, encapsulated in IP, encapsulated in Eth. To send frame to router, need MAC address of router interface: **ARP**
 - **ARP query** broadcast, received by router, which replies with **ARP reply** giving MAC address of router interface
-
- Client now knows MAC address of first hop router, so can now send frame containing DNS query

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A day in the life.... Using DNS

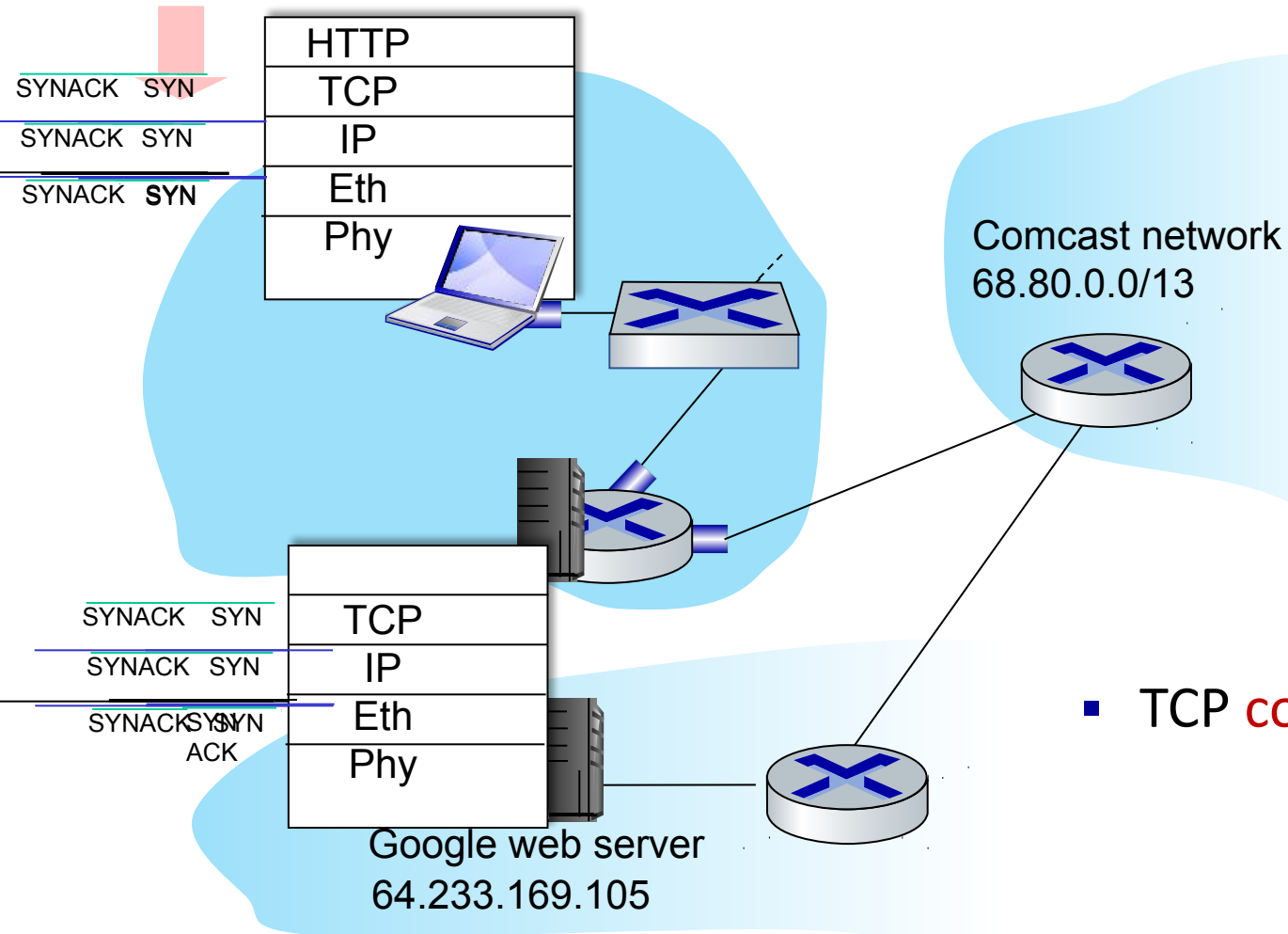


- IP datagram containing DNS query forwarded via LAN switch from client to 1st hop router
- IP datagram forwarded from campus network into Comcast network, routed (tables created by **RIP**, **OSPF**, **IS-IS** and/or **BGP** routing protocols) to DNS server

- Demuxed to DNS
- DNS replies to client with IP address of www.google.com

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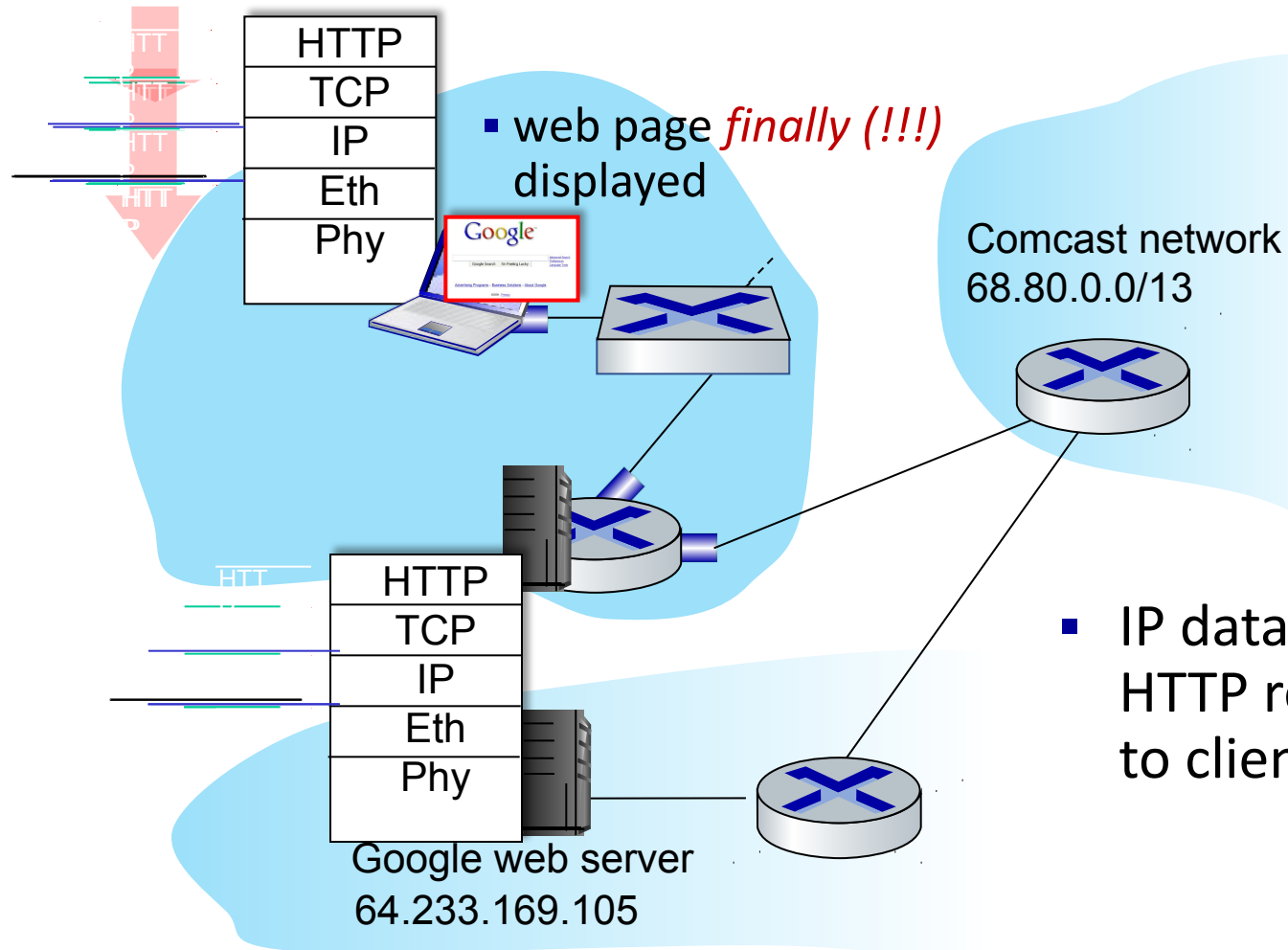
A day in the life.... TCP Connection carrying HTTP



- To send HTTP request, client first opens **TCP socket** to web server
- TCP **SYN segment** (step 1 in TCP 3-way handshake) inter-domain routed to web server
- Web server responds with **TCP SYNACK** (step 2 in TCP 3-way handshake)
- **TCP connection established!**

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A day in the life.... HTTP Request / Reply



- HTTP request sent into TCP socket
- IP datagram containing HTTP request routed to www.google.com
- Web server responds with HTTP reply (containing web page)
- IP datagram containing HTTP reply routed back to client



THANK YOU

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